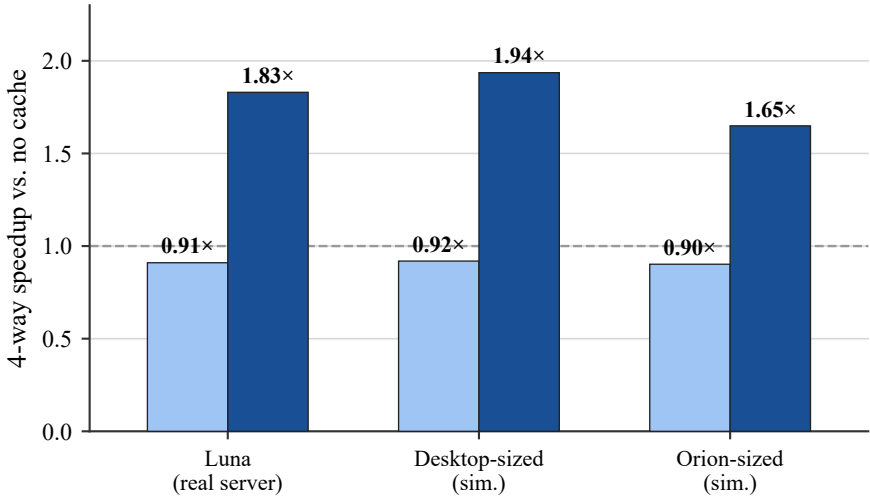


Cache benefit isn't simply
"smaller hardware cache = bigger win"

50 MB DB 8 GB DB



Level	Luna (real)	Desktop (sim.)	Orion (sim.)
L1 data	48KB / 12-way	32KB / 8-way	64KB / 4-way
L1 instr.	32KB / 8-way	32KB / 8-way	64KB / 4-way
L2	2MB / 16-way	512KB / 8-way	256KB / 8-way
LLC (total)	105MB / 15-way	32MB / 16-way	4MB / 16-way*
LLC shared by	96 cores	6 cores	12 cores
Clock	2.1 GHz	3.7 GHz	2.2 GHz

*Orion LLC associativity is an unverified placeholder (real value not publicly documented); all other values are real spec or sysfs-measured.

This chart is the reverse of the others: the SOFTWARE S2 cache is held FIXED at 4-way throughout - what changes is the HARDWARE cache config itself (table above). Desktop sees a stronger win than Luna's real cache; Orion sees a weaker one, despite having the smallest cache of the three. Config-only comparison (same x86 core model).